Sreenidhi Institute of Science and Technology

Because, Life is all about taking the right decisions

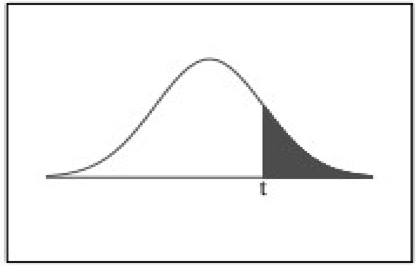
AUTONOMOUS

Lecture-2 **Unit-VI**

Tests of Hypothesis for Small Sample

K.RAMESH
Faculty of Mathematics
Department of Science & Humanities
SNIST

t-Distribution Table



The shaded area is equal to α for $t = t_\alpha$.

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
32	1.309	1.694	2.037	2.449	2.738
34	1.307	1.691	2.032	2.441	2.728
36	1.306	1.688	2.028	2.434	2.719
38	1.304	1.686	2.024	2.429	2.712
∞	1.282	1.645	1.960	2.326	2.576

t –test for difference of means

Let $\bar{x}, S_1^2$ and $\bar{y}, S_2^2$ be the mean and variance of two independent samples of sizes n_1 and n_2 ($n_1<30$ and $n_2<30$) drawn from two normal populations having the means μ_1 and μ_2. And population variances are variances σ_1^2 and σ_2^2 these are unknown and equal (i.e $\sigma_1 = \sigma_2$).

To test whether the two population means are equal.

Step1.Let N.H be H_0: $\mu_1 = \mu_2$

Step2.Let A.H be H_1: $\mu_1 \neq \mu_2$

Step3:LoS:α

Step.4:Apply two tailed test and calculate table value of t with n_1+n_2-2 dof.

(Note: If A.H is < type then apply L.O.T.T ,If A.H is > type then apply R.O.T.T)

Step.5:Test static $$t_{cal} = \frac{\bar{x} - \bar{y}}{S.\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where $\bar{x} = \frac{\sum_{i=1}^{n_1} x_i}{n_1}$ $\quad \bar{y} = \frac{\sum_{i=1}^{n_2} y_i}{n_2}$

$$S^2 = \frac{\sum(x_i - \bar{x})^2 + \sum(y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

Note: If S_1^2 and S_2^2 are given directly in the problem then use following formula

$$S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}$$

Step.6:Decision

Example1.

Samples of two types of electric light bulbs were tested for length of life and the following data were obtained

Type-I	Type-II
Sample number, $n_1=8$.	Sample number, $n_2=7$.
Sample mean, $\bar{x}=1234$ hours.	Sample mean, $\bar{y}=1036$ hours.
Sample S.D, $S_1=36$ hrs.	Sample S.D, $S_2=40$ hrs.

Is the difference in the means sufficient to warrant that type I is superior to type II regarding length of life?

Solution:

Given data

Type-I	Type-II
Sample number, $n_1=8$. Sample mean, $\bar{x}$ $=1234$ hours. Sample S.D, $S_1=36$ hrs.	Sample number, $n_2=7$. Sample mean, $\bar{y}$ $=1036$ hours. Sample S.D, $S_2=40$ hrs.

Null Hypothesis: $\mu_1 = \mu_2$

i.e., the two types I & type II of electric bulbs are same w.r.t length of life.

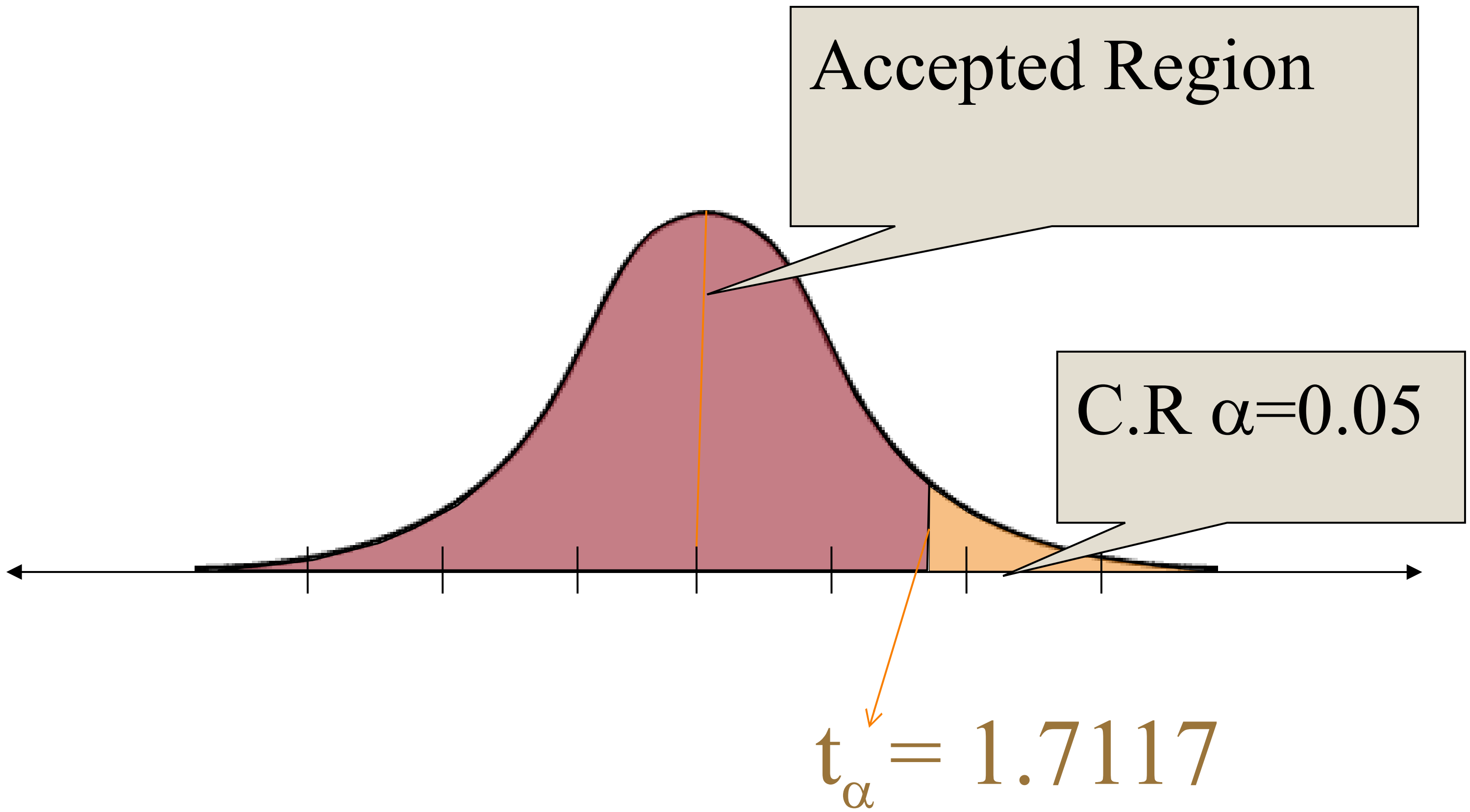
Alternative Hypothesis : $\mu_1 > \mu_2$

(Right tailed test)

LoS: $\alpha = 5\%$

C.R: $t_\alpha = t_{0.05} = 1.771$ **with 8+7-2=13 dof**

Right One Tailed Test (R.O.T.T)



$$S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}$$

$$S^2 = \frac{8(36)^2 + 7(40)^2}{8 + 7 - 2}$$

$$= 1659.06$$

$$t_{cal} = \frac{1234 - 1036}{\sqrt{1659.06\left(\frac{1}{8} + \frac{1}{7}\right)}} = 9.39$$

Decision: We reject N.H since $t_{cal} > t_{\alpha}$

That is the type I is superior than type II

Example.2:

Ten soldiers participated in a shooting competition in the first week. After intensive training they participated in the competition in the second week . Their scores before and after training are given below.

Scores before	67	24	57	55	63	54	56	68	33	43
Scores after	70	38	58	58	56	67	68	75	42	38

Do the data indicate that the soldiers have been benefited by the training.

Null Hypothesis: $\mu_1 = \mu_2$

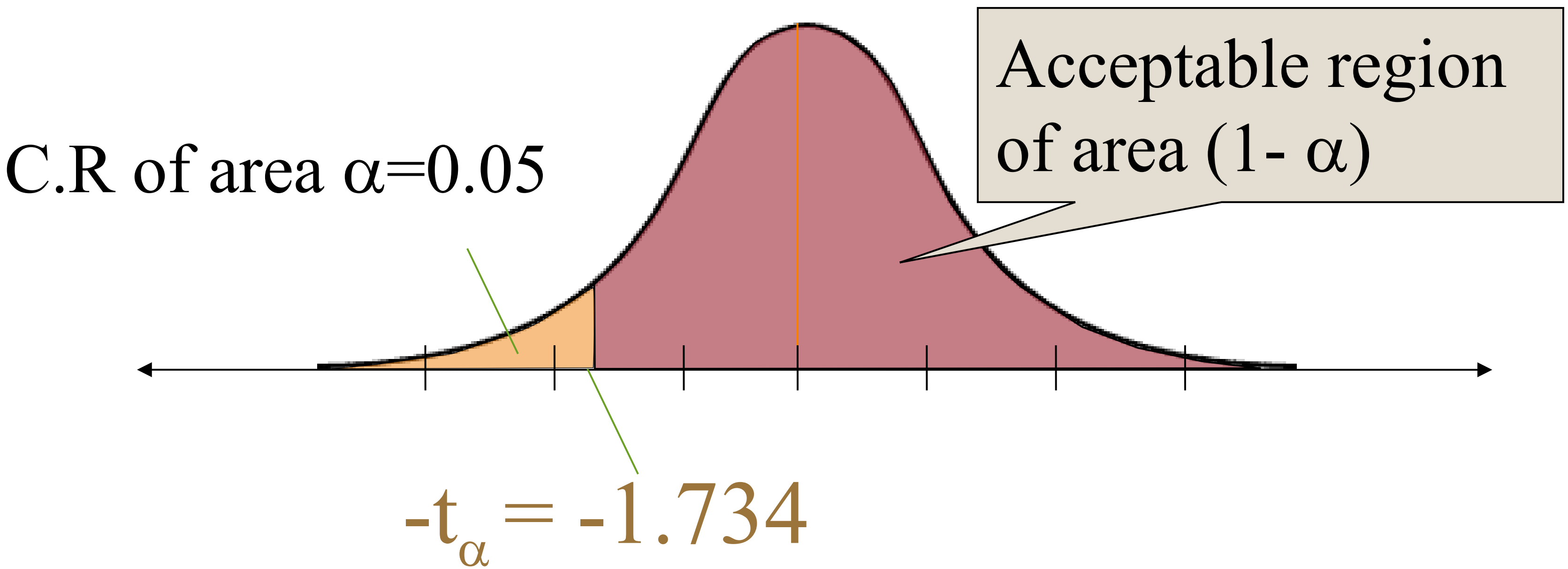
Alternative Hypothesis : $\mu_1 < \mu_2$ (LOTT)

(Right tailed test)

LoS: $\alpha = 5\%$

C.R: $-t_\alpha = -t_{0.05} = -1.734$ **with 10+10-2=18 dof.**

L.O.T.T


C.R of area α=0.05
Acceptable region of area (1- α)
$-t_\alpha = -1.734$

Test static

$$S^2 = \frac{1}{n_1+n_2-2}[(\sum(x - \ \overline{x})^2 + \sum(y - \ y)^2]$$

$$= \frac{1}{18}[1882 + 1664]$$

$$= \frac{1}{18}(3546)$$

$$= 197$$

$$\therefore S = \sqrt{197} = 14.0357$$

$$t_{cal} = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

$$= \frac{52 - 57}{14.0357\left(\sqrt{\frac{1}{10}+\frac{1}{10}}\right)}$$

$$\therefore\ t_{cal} = -0.796$$

Decision: We Accept N.H since $t_{cal} > -t_{\alpha}$

Example: 3

In a test given to two groups of students the marks obtained were as follows.

First Group	18	20	36	50	49	36	34	49	41
Second Group	29	28	26	35	30	44	46		

Examine the significant difference between the means of marks secured by students of the above two groups.

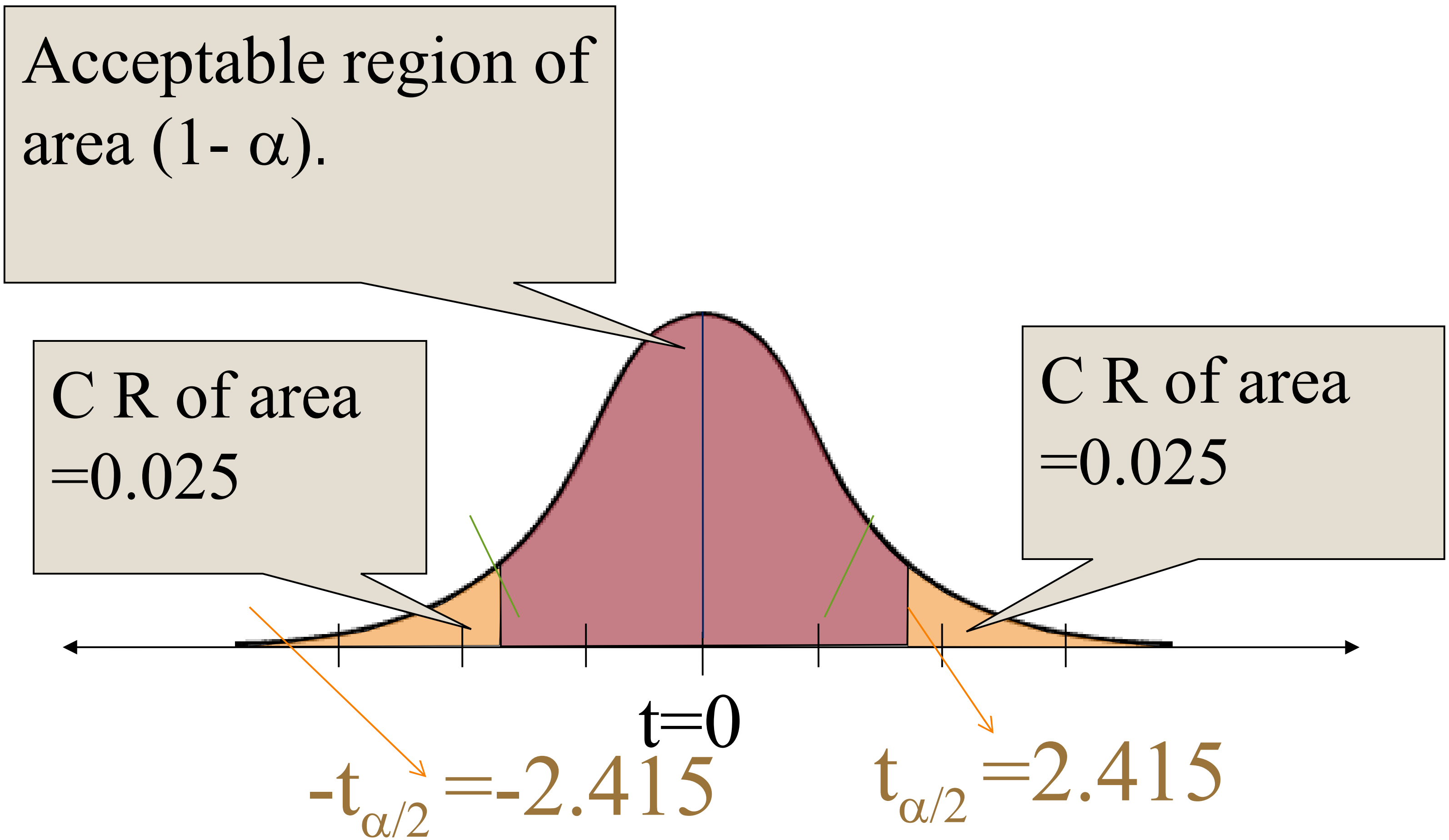
Solution:

Step1.Null Hypothesis:$\mu_1=\mu_2$

Step 2.Alternative Hypothesis : $\mu_1\neq\mu_2$

Step 3 LoS:α=5%

Step4.C.R: $t_{\alpha/2}=t_{0.05/2}=2.145$
with 9+7-2=14 dof


Acceptable region of area (1- α).
C R of area =0.025
C R of area =0.025
t=0
-tα/2 =-2.415
tα/2 =2.415

Step 5.Test static

x_i	18	20	36	50	49	36	34	49	41	$\sum x = 333$
$x_i - \bar{x}$	-19	-17	-1	13	12	-1	-3	12	4	
$(x_i - \bar{x})^2$	361	289	1	169	144	1	9	144	16	$\sum (x_i - \bar{x})^2 = 1134$
y_i	29	28	26	35	30	44	46	238		$\sum y = 238$
$y_i - \bar{y}$	-5	-6	-8	1	-4	10	12			
$(y_i - \bar{y})^2$	25	36	64	1	16	100	144	386		$\sum (y_i - \bar{y})^2 - 386$

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{333}{9} = 37$$

$$s_1^2 = \frac{\sum (x_i - \overline{x})^2}{n_1 - 1} = \frac{1134}{9 - 1} = 141.75$$

S_1=11.905

$$\bar{y} = \frac{\sum y_i}{n_2} = \frac{238}{7} = 34$$

$$s_2^2 = \frac{\sum (y_i - \overline{y})^2}{n_2 - 1} = \frac{386}{7 - 1} = 64.33$$

S_2=8.02

$$S^2 = \frac{9(141.75) + 7(64.33)}{9 + 7 - 2}$$
$$= 125.86$$

S=11.2

$$t_{cal} = \frac{37 - 34}{11.2\sqrt{\left(\frac{1}{9} + \frac{1}{7}\right)}} = 0.526$$

Decision: We Accept N.H since $t_{cal} \in (-t_{\alpha/2}, t_{\alpha/2})$ i.e value of t_{cal} falls in acceptance region.
Where ($t_{cal} = 0.526$, $t_{\alpha/2} = 2.145$)

That is there is no significant difference between the mean marks secured by the two groups.

Example : The nicotine content in milligrams of the samples of tobacco were found as follows:

Sample A	24	27	26	21	25	
Sample B	27	30	28	31	22	36

Can it be said that the two samples come from normal populations with the sample mean.

Solution: We have

$$\bar{x} = \frac{24 + 27 + 26 + 21 + 25}{5} = \frac{123}{5} = 24.6$$

$$\bar{y} = \frac{27 + 30 + 28 + 31 + 22 + 36}{6} = \frac{174}{6} = 29$$

Calculation of sample SD's

x	$x-\bar{x}$	$(x-\bar{x})^2$	y	$y-\bar{y}$	$(y-\bar{y})^2$
24	−0.6	0.36	27	−2	4
27	2.4	5.76	30	1	1
26	1.4	1.96	28	−1	1
21	−3.6	12.96	31	2	4
25	0.4	0.16	22	−7	49
–	–	–	36	7	49
123	0	21.2	174	0	108

$$\therefore \quad \sum (x_i - \overline{x})^2 = 21.2, \sum (y_i - \overline{y})^2 = 108$$

$$S^2 = \frac{1}{n_1 + n_2 - 2}\left[\sum_1^n (x_i - \overline{x})^2 + \sum_1^n (y_i - \overline{y})^2\right]$$

$$= \frac{1}{5 + 6 + 1}[21.2 + 108] = 14.35$$

$$\therefore \quad S = \sqrt{14.35} = 3.78$$

Null Hypothesis H_0: $\mu_1 = \mu_2$, i.e., the two samples have been drawn from normal populations with the same mean

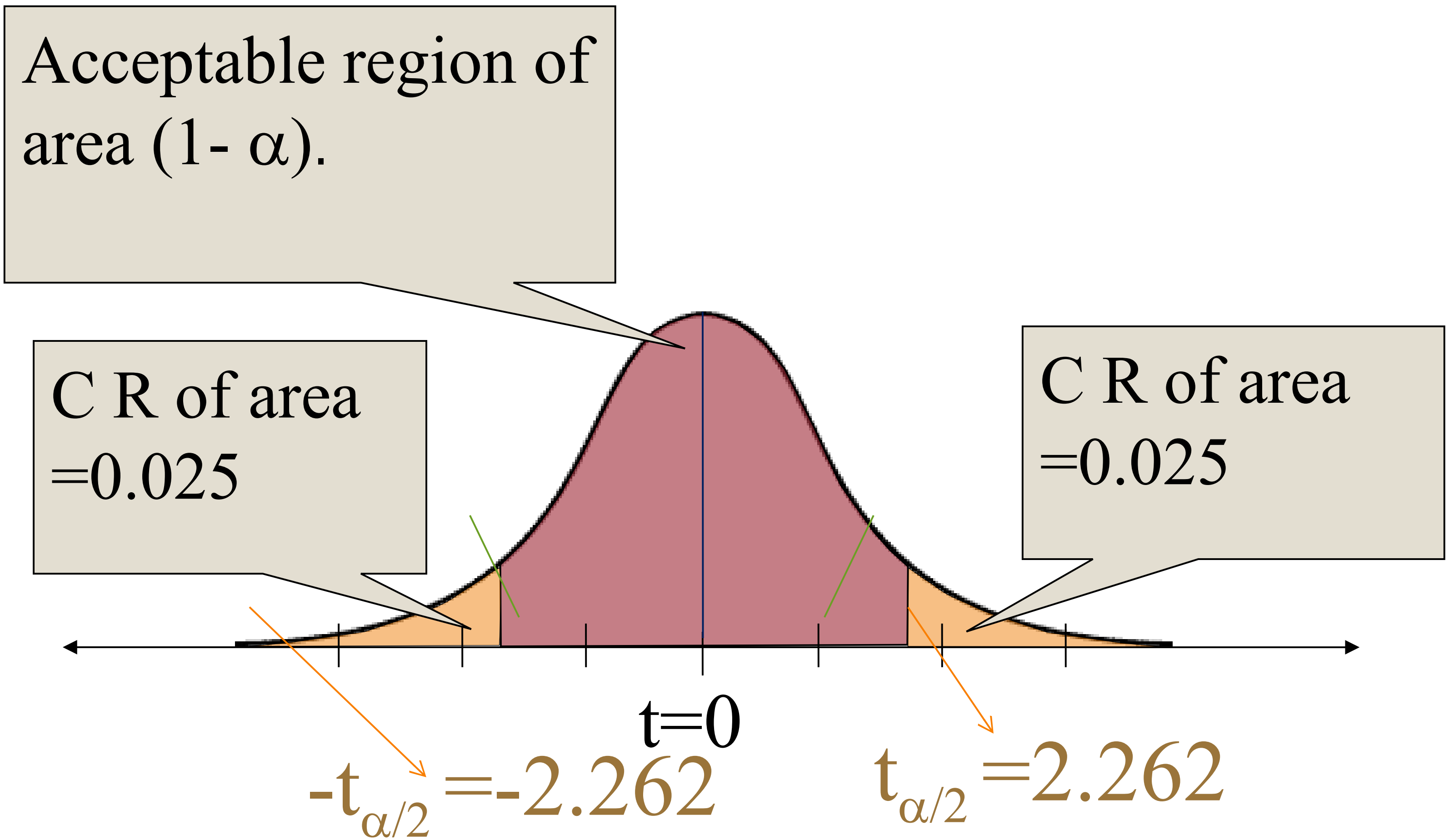
Alternative Hypothesis H_1: $\mu_1 \neq \mu_2$

Level of significance = 5%,

Degrees of freedom = $n_1 + n_2 - 2 = 5 + 6 - 2 = 9$

C.R: $t_{\alpha/2} = t_{0.05/2} = 2.262$

with 5+6-2=9 dof



Acceptable region of area (1- α).
C R of area =0.025
C R of area =0.025
t=0
-tα/2 =-2.262
tα/2 =2.262

$$t_{cal} = \frac{(\overline{x} - \overline{y})}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{24.6 - 29}{3.78\sqrt{\frac{1}{5} + \frac{1}{6}}} = -1.92$$

$$\therefore \qquad |t| = 1.92 < 2.262$$

Decision: We Accept N.H since $t_{cal} \in (-t_{\alpha/2}, t_{\alpha/2})$ i.e value of t_{cal} falls in acceptance region.

Where (t_{cal} =-1.92, $t_{\alpha/2}$ =2.262)

Thanks for watching this video

Lecture-3

Unit-VI

Tests of Hypothesis for Small Sample

K.RAMESH
Faculty of Mathematics
Department of Science & Humanities
SNIST

F-Distribution :

The F-Distribution is also called as Variance Ratio Distribution.

A random variable F is said to follow the F-distribution with (ν_1, ν_2) degrees of freedom, if its probability density is given by

$$f(F) = \frac{\left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} F^{\left(\frac{\nu_1}{2}\right)-1}}{\beta\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)\left[1+\left(\frac{\nu_1}{\nu_2}\right)F\right]^{\frac{(\nu_1+\nu_2)}{2}}}, \quad F > 0$$

We usually denote $F \backsim F(\nu_1, \nu_2)$.

Let $x_1, x_2, \dots x_{n1}$ be random sample drawn from a normal population with mean μ_1 and Variance σ_1^2 and let $y_1, y_2, \dots y_{n2}$ be random sample drawn from a normal population with mean μ_2 and Variance σ_2^2. Assume that both normal populations are independent.

Let S_1^2 and S_2^2 be the sample variances of first and second samples respectively. Then the ratio

$$F_{cal} = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

Which follows F-distribution with $\upsilon_1 = n_1 - 1$ numerator degrees of freedom and $\upsilon_2 = n_2 - 1$ denominator degrees of freedom.

Properties of F-distribution:

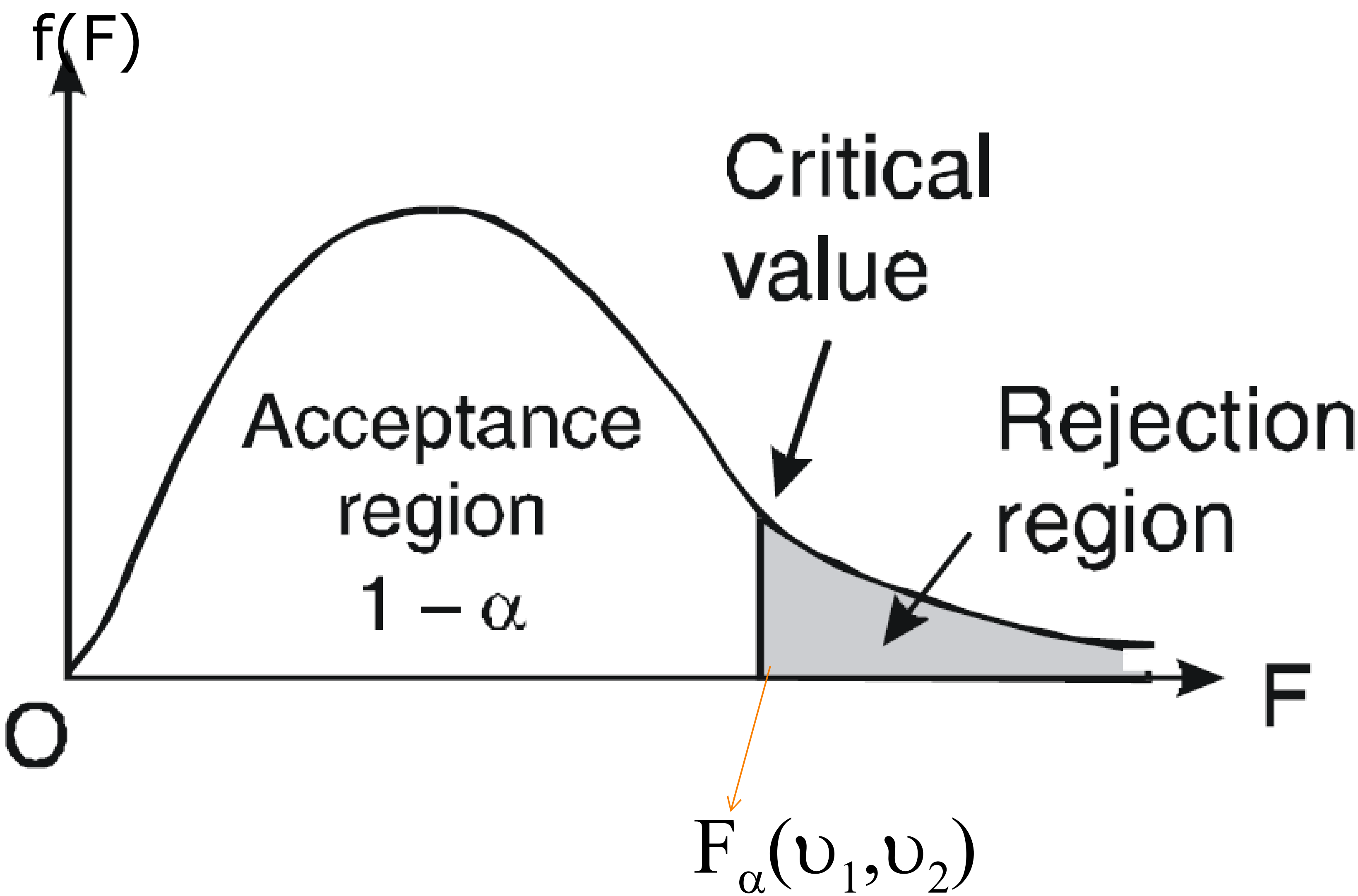
1. The shape of the F-distribution is free from population parameters and **depends on its parameters** υ_1 and υ_2 degrees of freedom.

2. F-distribution curve lies entirely lies in first-quadrant and is unimodal.

3. The F-curve depends not only on the two parameters υ_1 and υ_2 but also on the order in which they are stated.

4. $$F_{(1-\alpha)}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$$

Note: $F_{\alpha}(v_1, v_2)$ is the value of F with v_1 and v_2 degrees of freedom such that the area under the F-distribution curve to the right of F_{α} is α


f(F)
Critical
value
Acceptance
region
1 – α
Rejection
region
O
F
$F_{\alpha}(\upsilon_1,\upsilon_2)$

F Distribution: Critical Values of *F* (5% significance level)

v_1 / v_2	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.36	246.46	247.32	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.42	19.43	19.44	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.71	8.69	8.67	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.87	5.84	5.82	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.64	4.60	4.58	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.96	3.92	3.90	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.53	3.49	3.47	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.24	3.20	3.17	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.03	2.99	2.96	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.86	2.83	2.80	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.74	2.70	2.67	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.64	2.60	2.57	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.55	2.51	2.48	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.48	2.44	2.41	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.42	2.38	2.35	2.33

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.37	2.33	2.30	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.33	2.29	2.26	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.29	2.25	2.22	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.26	2.21	2.18	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.22	2.18	2.15	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.20	2.16	2.12	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.17	2.13	2.10	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.15	2.11	2.08	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.13	2.09	2.05	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.11	2.07	2.04	2.01
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.09	2.05	2.02	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.08	2.04	2.00	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.06	2.02	1.99	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.05	2.01	1.97	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.04	1.99	1.96	1.93
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11	2.04	1.99	1.94	1.91	1.88
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.95	1.90	1.87	1.84
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.95	1.89	1.85	1.81	1.78
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.86	1.82	1.78	1.75

F Distribution: Critical Values of *F* (1% significance level)

v_2 \ v_1	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
1	4052.18	4999.50	5403.35	5624.58	5763.65	5858.99	5928.36	5981.07	6022.47	6055.85	6106.32	6142.67	6170.10	6191.53	6208.73
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.44	99.44	99.45
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.92	26.83	26.75	26.69
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.25	14.15	14.08	14.02
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.77	9.68	9.61	9.55
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.60	7.52	7.45	7.40
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.36	6.28	6.21	6.16
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.56	5.48	5.41	5.36
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	5.01	4.92	4.86	4.81
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.60	4.52	4.46	4.41
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.29	4.21	4.15	4.10
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.05	3.97	3.91	3.86
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.86	3.78	3.72	3.66
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.70	3.62	3.56	3.51
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.56	3.49	3.42	3.37

F-test is used to test

1. Whether two independent samples have been drawn from normal populations with the same variance. F-test is extremely useful for comparing variances of two populations.

2. Whether two independent estimates of the population variances are homogeneous or not.

Snedecor's F- Test of Significance

Test for equality of two population variances:

Step1: N.H: $H_0: \sigma_1^2 = \sigma_2^2$

Step2: A.H: $H_1: \sigma_1^2 \neq \sigma_2^2$

Step3: Write LoS: α

Step4: Find $F_{cal} = \frac{S_1^2}{S_2^2}$ where $S_1^2 > S_2^2$

i.e $$F_{cal} = \frac{\textbf{Greater Variance}}{\textbf{Smaller Varinace}}$$

where

$$S_1^2 = \sum_{i=1}^{n_1} \frac{(x_i - \overline{x_\iota})^2}{n_1 - 1} \qquad S_2^2 = \sum_{i=1}^{n_2} \frac{(y_i - \overline{y_\iota})^2}{n_2 - 1}$$

Step5: Find $F_\alpha(\upsilon_1, \upsilon_2)$ from the table where $\upsilon_1 = n_1 - 1$,

$\upsilon_2 = n_2 - 1$ dof.

Step6.Decision: Accept N.H if $F_{cal} < F_{\alpha}$

Example1.

Two random samples are

Sample	*Size*	*Sum of squares of deviations from the mean*
1	*10*	*90*
2	*12*	*108*

Test whether the samples come from normal population at 5% level of significance.

Solution.

Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

The two samples have been drawn from the same normal population

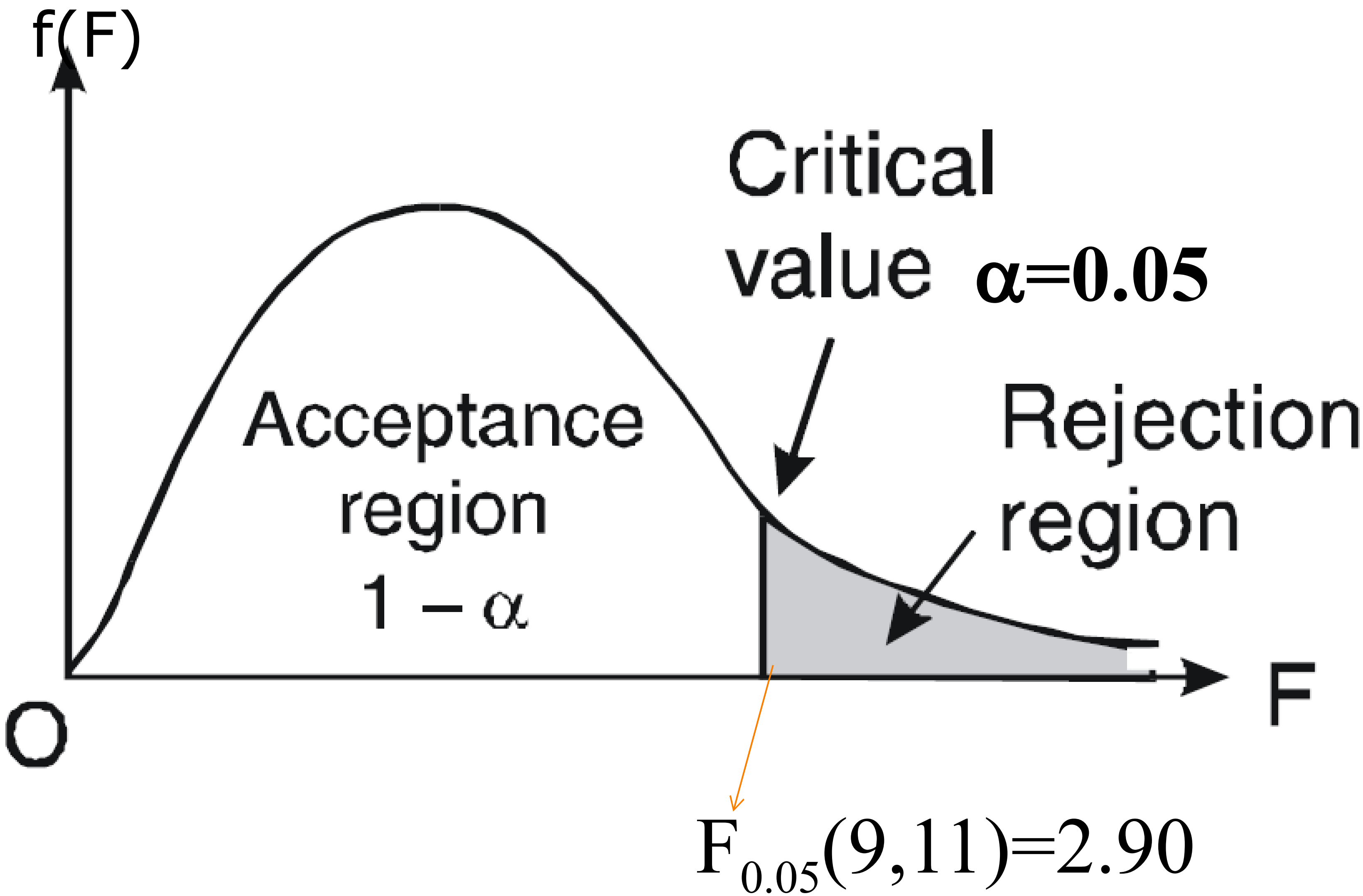
Alternate Hypothes $H_1: \sigma_1^2 \neq \sigma_2^2$

The two samples have been drawn from the normal populations with different variance.

LoS: $\alpha = 5\%$

C.R: Given $n_1 = 10$ $n_2 = 12$

Tabulated $F_{0.05}\,(9, 11) = 2.90$


f(F)
Critical
value
α=0.05
Acceptance
region
1 − α
Rejection
region
O
F
F0.05(9,11)=2.90

Test static

$$n_1 = 10,\ \sum(x-\bar{x})^2 = 90$$

$$n_2 = 12,\ \sum(y-\bar{y})^2 = 108$$

$$S_1^2 = \frac{1}{n_1-1}\sum(x-\bar{x})^2 = \frac{90}{9} = 10$$

$$S_2^2 = \frac{1}{n_2-1}\sum(y-\bar{y})^2 = \frac{108}{11} = 9.82$$

$$F_{cal} = \frac{S_1^2}{S_2^2} \quad \text{since } S_1^2 > S_2^2$$

$$= \frac{10}{9.82} = 1.018$$

Decision: Since the calculated F is less than tabulated F, it is not significant. Hence, null hypothesis of equality of population variances is accepted.

Ex pumpkins were grown under two Experimental Conditions. Two random samples of 11 and 9 pumpkins, show the sample standard deviations of their weights as 0.8 and 0.5, respectively. Test the hypothesis that the true variances are equal.

Assuming that the weight are
distributed normally

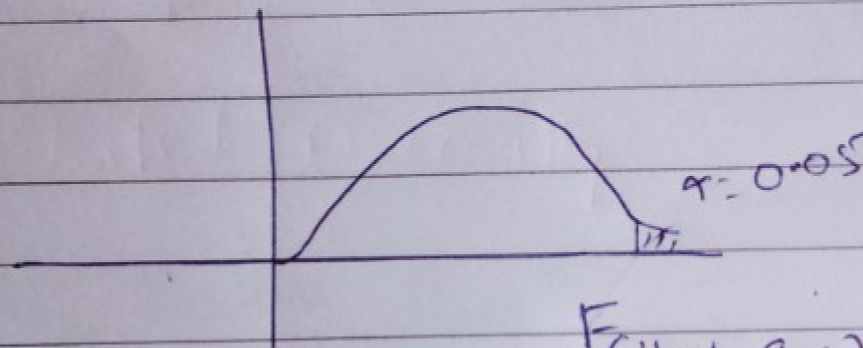
Note: Samples variances are given directly.

Soln Given $n_1 = 11$, $n_2 = 9$

Step 1: Let N.H be $H_0: \sigma_1^2 = \sigma_2^2$

Step 2: Let A.H be $H_1: \sigma_1^2 \neq \sigma_2^2$

Step 4: Critical Region:



$$F_{\alpha(11-1,\ 9-1)}$$

$$F_{0.05}(10, 8) = 3.35$$

where $\nu_1 = n_1 - 1 = 10$

$\nu_2 = n_2 - 1 = 8$

Date :

Page No.

Step 5 : Test static

If S_1^2 and S_2^2 be the estimates

of σ_1^2 and σ_2^2 then

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{11 \times (0.8)^2}{10} = 0.704$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{9 \times (0.5)^2}{8} = 0.281$$

Test Static $F_{cal} = \frac{S_1^2}{S_2^2} \quad (\because S_1^2 > S_2^2)$

$$= \frac{0.704}{0.281}$$

$$= 2.5$$

Step 6: Decision: Accept N.H at 5%
LOS.

Conclusion: The variance of two populations is same.

∴ The two samples have same variance.

Ex Two random samples reveal the following results.

Sample	Size	Sample Mean	Sum of squares of Deviations from the Mean
1	10	15	90
2	12	14	108

Test whether the samples came from same normal population.

Soln

F - test:

Step 1: $\sigma_1^2 = \sigma_2^2$

Step 2 : $\sigma_1^2 \neq \sigma_2^2$

Step 3 : LOS $\alpha = 5\%$

Step 4 : C.R $F_{\alpha}(\nu_1, \nu_2) = F_{0.05}(9, 11) = 2.90$

Step 5:

Given $n_1 = 10$, $n_2 = 12$,

$\bar{x}_1 = 15$ $\quad \bar{x}_2 = 14$

$\Sigma(x_i - \bar{x})^2 = 90$ $\quad \Sigma(y_i - \bar{y})^2 = 108$

$$s_1^2 = \frac{\Sigma(x_i - \bar{x})^2}{n_1 - 1} = \frac{90}{9} = 10$$

$$S_2^2 = \frac{\Sigma (y_i - \bar{y})^2}{n_2 - 1} = \frac{108}{11} = 9.82$$

$$F_{cal} = \frac{S_1^2}{S_2^2} = \frac{10}{9.82} = 1.018$$

Step 6:

Decision: Accept N.H. since

$F_{cal} < F_{tab}$

∴ The samples came from same normal population.

(OR)

The samples came from populations with same variance.

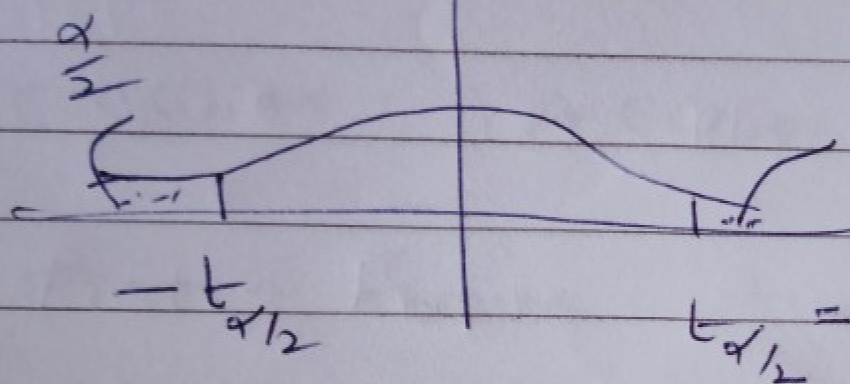
t - test:

N.H: $\mu_1 = \mu_2$

A.H $\mu_1 \neq \mu_2$

LOS: $\alpha = 5\%$

critical region: Apply two tailed test



$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$

$t_{0.025} = 2.086$

with $\nu = 10 + 12 - 2 = 20$

Test Static:

$$s^2 = \frac{1}{n_1+n_2-2}\left[\Sigma(x_i-\bar{x})^2 + \Sigma(y_i-\bar{y})^2\right]$$

$$= \frac{1}{10+12-2}\left[90+108\right] = 9.9$$

$$s = 3.15$$

$$t_{cal} = \frac{\bar{x}-\bar{y}}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{15-14}{3.15\sqrt{\frac{1}{10}+\frac{1}{12}}} = 0.74$$

Decision : Accept N.H.

From above two tests, the given samples drawn from same normal population.

Hence we accept null hypothesis

$\mu_1 = \mu_2$ and $\sigma_1^2 = \sigma_2^2$

Example : Two horses A and B were tested according to the time (in seconds) to run on a particular track with the following results:

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	–

Test whether the two horses have the same running capacity?

Solution: We have $n_1 = 7$, $n_2 = 6$

Then

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{219}{7} = 31.28$$

$$\bar{y} = \frac{\sum y_i}{n_2} = \frac{169}{6} = 28.2$$

Let the time taken by horse A be denoted by x and the time taken by horse B be denoted by y

Degrees of freedom = $n_1 - 1$, $n_2 - 1 = 6, 5$

Critical value, i.e., level of significance = 5%

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$y - \bar{y}$	$(y - \bar{y})^2$
28	− 3.28	10.75	28	0.	0.64
30	− 1.28	1.64	30	1.8	3.24
32	0.72	0.52	30	1.8	3.24
33	1.72	2.96	24	− 4.2	17.64
33	1.72	2.96	27	− 1.2	1.44
29	− 2.28	5.20	29	0.8	0.64
34	2.72	7.40	—	—	—
219		31.43	169		26.84

$$S_1^2 = \frac{1}{n_1 - 1}\sum (x - \overline{x})^2 = \frac{31.43}{6} = 5.224$$

$$S_2^2 = \frac{1}{n_2 - 1}\sum (y - \overline{y})^2 = \frac{26.84}{5} = 5.368$$

Null Hypothesis H_0: The two horses have the same running capacity, i.e., $\sigma_1^2 = \sigma_2^2$

Alternative Hypothesis H_1: $\sigma_1^2 \neq \sigma_2^2$

Table value of F for (5, 6) degrees of freedom at 5% level of significance is 4.39.

Test statistic is $F = \dfrac{S_2^2}{S_1^2} = \dfrac{5.368}{5.224} = 1.02 \; (\text{since } S_2^2 > S_1^2)$

Since the calculated value of F is less than the table value of F for (5, 6) df at 5% level of significance, we accept H_0.

The two horses have the same running capacity.

Thanks for watching this video

Rough

The sampling distribution of F is of the form

$$f(F) = \mathrm{K} \frac{F^{\frac{(\nu_1 - 2)}{2}}}{(\nu_1 F + \nu_2)^{\frac{(\nu_1 + \nu_2)}{2}}}$$

Where ν_1 and ν_2 are degrees of freedom and K is

determined by $\int_0^\infty f(F)dF = 1$.